



Assignment

Programming Fundamentals

Question 1:

To use ROT13, you simply replace each letter in a message with the letter at the 13 positions after it in the alphabets. For example, the message "HELLO" would be translated to "URYYB" in ROT13. Write a program that takes a character input from the user and finds out its corresponding ROT13 code. The program must display the resultant character.

Inputs:

A character

Outputs:

Corresponding ROT13 code of character

Computation:

Replace each letter with the letter that is at the 13th location of the original character. No character should be replaced with an invalid character. For example, you cannot replace 'Z' with any character as there is no letter after it. So 'Z' will be translated to 'M' which is at the 13th location counting back from 'Z'.

Question 2

Write a C++ program using functional decomposition that calculates a user's weight on a planet based on their input.

Inputs:

User's weight (in pounds)

Name of a planet

Outputs:

User's weight on the specified planet (in pounds)

Computation:

Factors by which the weight must be multiplied for each planet:

- Mercury: 0.4155
- Venus: 0.8975
- Earth: 1.0
- Moon: 0.166
- Mars: 0.3507
- Jupiter: 2.5374
- Saturn: 1.0677
- Uranus: 0.8947
- Neptune: 1.1794
- Pluto: 0.0899

The program should output an error message if the numbers entered are not in the required ranges. The prompt and the error message should make it clear to the user how the numbers must be entered. Be sure to use proper formatting and appropriate comments in your code. The output should be labeled clearly and formatted neatly.

Question 3

Write a C++ program using functional decomposition that determines the name of the day of the week based on a starting day of the year and a day of the year input.

Inputs:

Starting day of the year, represented by a number in the range of 0 to 6 (0 being Sunday, 1 being Monday, and so on)

Day of the year, represented by a number in the range of 1 to 366

Outputs:

Name of the day of the week corresponding to the day of the year

Computation:

The program should compute the number of the day of the week using the formula:

$$(\text{start day} + \text{day of year} - 1) \% 7$$

The program should output an error message if the numbers entered are not in the required ranges. The error message should clearly explain how the numbers must be entered.

Question 4

Write a C++ program that calculates the score for a bowler's three throws in a game of bowling.

Inputs:

Number of pins knocked down by the bowler in each of their three throws

Outputs:

Computed score for the three throws

Computation:

If the first throw is a strike (all 10 pins knocked down), the score is equal to 10 plus the number of pins knocked down in the next two throws.

If the first throw knocks down fewer than 10 pins, but the second throw knocks down the remainder of the 10 pins (a spare), the score is equal to 10 plus the number of pins knocked down in the third throw.

If the first two throws fail to knock down all of the pins (a blow), the score is just the total number of pins knocked down in the first two throws.

The maximum score for three strikes is 30.

Question 5

To create a C program that calculates a Qirat competition score, you can utilize functional decomposition.

Input:

The program should take input from four judges who assign scores between 0 and 10 to each Qari.

Output:

Display competition score for each of the qari.

Computation:

To compute the final score, the program should disregard the lowest score and average the three highest scores. However, if any of the scores are outside the 0 to 10 range, the program should output an error message instead of the final score.

Question 6

Write a program that asks the user to enter the number of pennies, nickels, and dimes required to make exactly one dollar. The program should then check if the total value of the coins entered is equal to one dollar. If it is, the program should print a message congratulating the user for winning the game. If the total value is less than one dollar, the program should print a message indicating how much less it is. If the total value is more than one dollar, the program should print a message indicating how much more it is.

Input:

Number of pennies, nickels, and dimes,

Output:

Amount in dollars composed of the coins.

Computation:

To solve this problem, you will need to define the value of each coin in cents (1 penny = 1 cent, 1 nickel = 5 cents, 1 dime = 10 cents, 1 quarter = 25 cents). There are 100 cents in each dollar.

Question 7:

Input:

The program should prompt the user to enter the number of units consumed by the customer.

Output:

The program should display the amount due for the customer.

Computation:

The program should calculate the electricity charges based on the following criteria:

- If the units consumed are between 1-100, the electricity charge per unit is Rs. 2.00.
- If the units consumed are between 101-200, the electricity charge per unit is Rs. 3.50.
- If the units consumed are 200 or more, the electricity charge per unit is Rs. 4.50.

General sales tax at a rate of 10% is also deducted from each consumer.